

Introduction to Microeconomics

Lecture 5 & 6—Elasticity

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x-elasticity of y

Consider a functional relationship between an endogenous variable y and an exogenous variable x , given by

$$y = f(x).$$

Then the **x-elasticity of y** (or elasticity of y with respect to x) is given by

$$\epsilon = \frac{\Delta\%y}{\Delta\%x} = \frac{\frac{\Delta y}{y}}{\frac{\Delta x}{x}} = \left(\frac{\Delta y}{\Delta x} \right) \left(\frac{x}{y} \right).$$

$\frac{\Delta y}{\Delta x}$ = rate of change.

Interpretation: The percentage change in variable y resulting from a 1% increase in variable x .

Price Elasticity of Demand

Price Elasticity of Demand

Percentage change in quantity demanded of a good resulting from a 1% increase in its **own price**.

$$\epsilon_p^D = \frac{\Delta\%Q}{\Delta\%p} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta p}{p}} = \left(\frac{\Delta Q}{\Delta p}\right) \left(\frac{p}{Q}\right).$$

Example

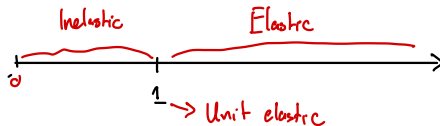
If a 1% rise in the price of wheat caused a 2% reduction in the quantity of wheat demanded, the price elasticity of demand for wheat would be -2 . *How?* $\frac{-2}{1} = -2$

Attention!

The price elasticity of demand will always be non-positive because price changes always move in the opposite direction from changes in quantity demanded (the law of demand).

Dictionary of terminology and conventions

- Out of convenience, we usually drop the negative sign and speak of price elasticities in absolute values.
- The demand for a good is said to be:
 - **Elastic** with respect to its price if its price elasticity is more than 1. $|\epsilon_p^D| > 1$
 - **Inelastic** with respect to its price if its price elasticity is less than 1. $|\epsilon_p^D| < 1$
 - **Unitary Elastic** with respect to price if its price elasticity is equal to 1. $|\epsilon_p^D| = 1$



Price Elasticity of Demand

Recall the formula

$$\epsilon_p^D = \underbrace{\left(\frac{\Delta Q}{\Delta p} \right)}_{(1)} \underbrace{\left(\frac{p}{Q} \right)}_{(2)}.$$

$$\ast \lim_{\Delta P \rightarrow 0} \frac{\Delta Q}{\Delta P} = \frac{dQ}{dP}$$

↳ derivative of $Q^D(p)$ w.r.t. p

It has two components:

- ① $\frac{\Delta Q}{\Delta p}$, needs to be computed from the demand function.
- ② $\frac{p}{Q}$, needs a specific point (Q, p) in the demand curve.

When in (2) above you pick a specific point (Q, p) to compute the elasticity, we call it the **point-price elasticity of demand**.

A difficulty remains, how to compute (1)? You have two choices:

- You need a second point to compute an approximation of term (1)
- You need full information about the shape of the demand function $Q_d(p)$ and use Calculus to compute the derivative with respect to price. $\ast$

$$\begin{matrix} (Q^1, p^1) \\ (Q^0, p^0) \end{matrix} \left\} \frac{\Delta Q}{\Delta P} = \frac{Q^1 - Q^0}{p^1 - p^0}$$

Price elasticity for linear demand curves

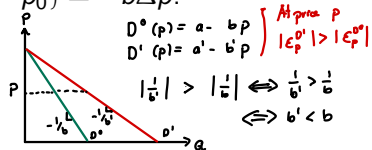
$$\begin{matrix} a > 0 \\ b > 0 \end{matrix}$$

Suppose that $Q_d(p) = a - bp$; and suppose that $\Delta p = p_1 - p_0$. Then

$$\Delta Q = Q_1 - Q_0 = (\cancel{a} - bp_1) - (\cancel{a} - bp_0) = -b(p_1 - p_0) = -b\Delta p.$$

Then

$$\frac{\Delta Q}{\Delta p} = -b.$$



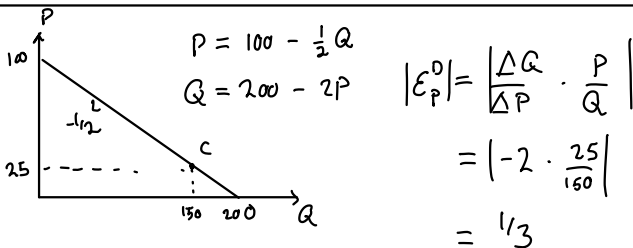
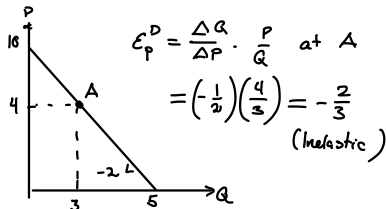
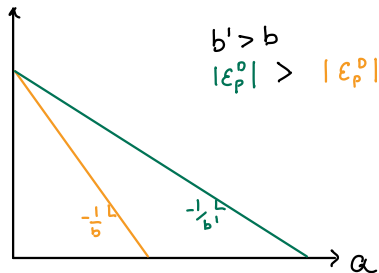
Recall that when graphing Q_d , we graph the inverse demand curve $p = a/b - \frac{1}{b}Q_d$.

Thus, for a linear demand curve, the **point-price elasticity of demand at point (Q, p)** is

$$\epsilon_p^D = -b \frac{p}{Q} = \left(\frac{\Delta Q}{\Delta P} \right) \frac{p}{Q} = -b \cdot \frac{p}{Q} = -b \frac{p}{a - bp}$$

Notice that the elasticity is inversely proportional to the slope of the inverse demand curve.

Examples



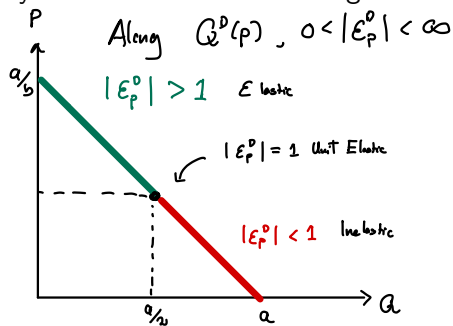
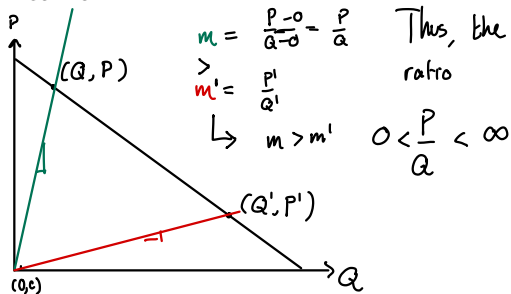
at this (25, 150) the demand is inelastic.

Discussion

- Notice that in the expression for the point-price elasticity of a linear demand curve

$$\epsilon_p^D = -b \frac{P}{Q} = -b \frac{P}{a - bP}$$

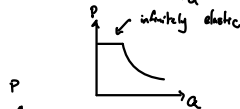
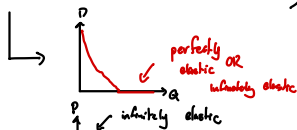
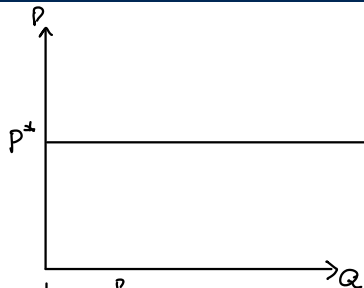
b is constant, but the ratio P/Q is not, it actually declines as we move along the demand curve.



Infinitely Elastic Demand

- For a horizontal demand curve $\Delta Q/\Delta p$ is infinite.
- A tiny change in price triggers an enormous change in demand $\Rightarrow$ the elasticity of demand is infinite.
- Also called **Perfectly Elastic Demand**.
- Intuition: consumers will buy as much of a good as they can get at a single price, but for any higher price the quantity demanded drops to zero, while for any lower price the quantity demanded increases without limit.

$$|\epsilon_p^D| \rightarrow \infty$$

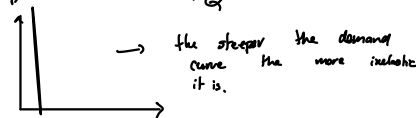
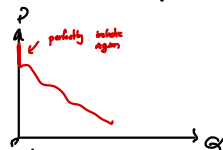
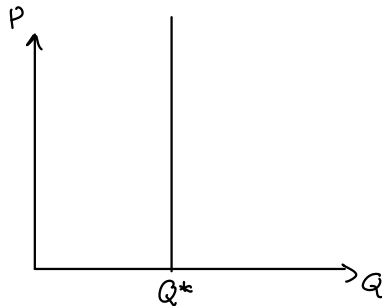


$\rightarrow$ the flatter the demand curve is the more elastic it is.

Infinitely Inelastic Demand

$$|\epsilon_p^p| = 0$$

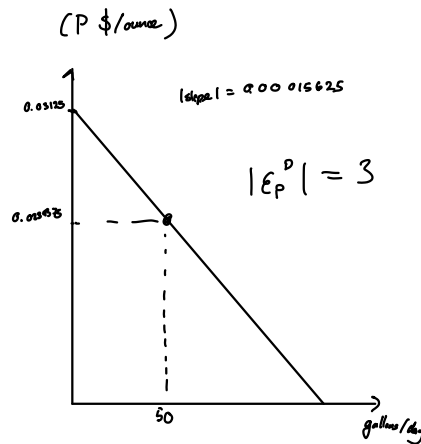
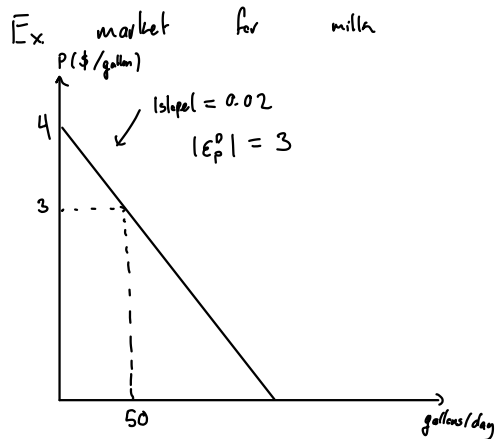
- For a vertical demand curve $\Delta Q / \Delta p$ is 0.
- Because the quantity demanded is the same no matter what the price, the elasticity of demand is zero.
- Also called **Perfectly Inelastic Demand**.
- Intuition: Consumers will buy a fixed quantity of a good, regardless of its price.



The unit-free property of Elasticity

Why measure price-responsiveness with elasticity, which seems complex, rather than with the slope, which seems simple?

Elasticity is unit-free, whereas the slope is not.



Elasticity and Total Expenditure or Revenue

Case 1

Suppose that you are the administrator of the toll booths at a highway. At the current toll of \$1/trip, 100,000 trips per hour are taken across the highway. If the price elasticity of demand for trips is (in absolute value) 2, what will happen to the number of trips taken per hour if you raise the toll by 10%? What happens to total expenditures?

$$\Delta\%P = 10\% \Rightarrow \Delta\%Q = -2 \cdot (10\%) = -20\% \rightarrow 80,000 \text{ trips/hour}$$

$$|E_P^D| = 2$$

$$\text{Original Revenue} = 100,000$$

$$\text{New Revenue} = (80,000) (\$1.10) = \$88,000$$

Recall $|E_P^D| = 2$, Demand is elastic

↳ When demand is elastic } $\uparrow P \Rightarrow \downarrow \text{Revenue}$

Case 2

Now suppose that the price elasticity is 0.5. How would the number of trips and total revenues then be affected by a 10% increase in the toll?

$$\Delta\%P = 10\% \Rightarrow \Delta\%Q = -0.5(10\%) = -5\% \rightarrow 95,000 \text{ trips/hour}$$

$$\text{New Revenue} = (95,000) \cdot (\$1.10) = \$104,500$$

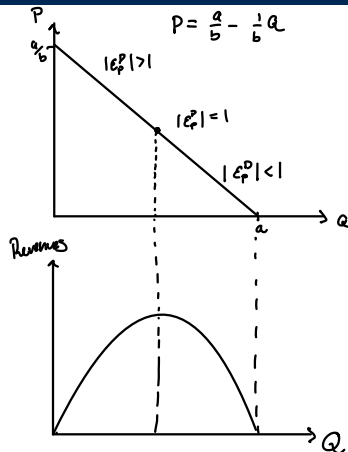
$|E_P^D| = 0.5$, Demand is inelastic

when demand is inelastic } $\uparrow P \Rightarrow \uparrow \text{Revenue}$

Elasticity and Total Expenditure or Revenue

- A price reduction will increase total revenue if and only if the price elasticity of demand is greater than 1.
- An increase in price will increase total revenue if and only if the price elasticity of demand is less than 1.

If demand is	$P \uparrow$	$P \downarrow$
Elastic $ \epsilon_P > 1$	$\downarrow R$	$\uparrow R$
Inelastic $ \epsilon_P < 1$	$\uparrow R$	$\downarrow R$



$$\begin{aligned}
 R &= P \cdot Q \\
 &= P \cdot Q(P) \\
 &= P \cdot (a - bP) \\
 \text{also } &= \left(\frac{a}{b} - \frac{1}{b}Q\right)Q \\
 &= \frac{a}{b}Q - \frac{1}{b}Q^2
 \end{aligned}$$

$$\begin{aligned}
 R &= P \cdot Q(P) \\
 \frac{dR}{dP} &= P \cdot \frac{dQ(P)}{dP} + (1)Q(P)
 \end{aligned}$$

$\rightarrow$ P^* at which R is maximized where $Q^* = Q(P)$
 $\left(\frac{1}{Q^*}\right)P^* + \frac{dQ(P^*)}{dP} + Q(P^*)\left(\frac{1}{Q^*}\right) = 0$
 $\left(\frac{P^*}{Q^*}\right)\frac{dQ(P^*)}{dP} + 1 = 0 \Rightarrow |\epsilon_P^D| = 1$

Determinants of the price elasticity of demand

- 1 Substitution Possibilities
- 2 Budget share.
- 3 Time.

Other demand elasticities with respect to its determinants

Income Elasticity of Demand

Percentage change in the quantity demanded resulting from a 1% increase in income.

$$\epsilon_I^D = \frac{\Delta\%Q}{\Delta\%I} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta I}{I}} = \left(\frac{\Delta Q}{\Delta I} \right) \left(\frac{I}{Q} \right).$$

Cross-price elasticity of demand

Percentage change in the quantity demanded of one good resulting from a 1% increase in the price of another.

$$\epsilon_{p_i}^{D_j} = \frac{\Delta\%Q_j}{\Delta\%p_i} = \frac{\frac{\Delta Q_j}{Q_j}}{\frac{\Delta p_i}{p_i}} = \left(\frac{\Delta Q_j}{\Delta p_i} \right) \left(\frac{p_i}{Q_j} \right).$$

Dictionary of terminology and conventions

- For the income elasticity, we say the good is
 - **Inferior** with respect to income if its income elasticity is negative.
 - **Independent** with respect to income if its income elasticity is zero.
 - **Normal** with respect to income if its income elasticity is positive.
 - Some authors/textbooks call a good **Superior** with respect to income if its income elasticity is greater than one.
- For the cross-price elasticity of good Q_j with respect to the price of good Q_i we say they are:
 - **Complements** if the cross-price elasticity of j with respect to the price of i is negative.
 - **Substitutes** if the cross-price elasticity of j with respect to the price of i is positive.
 - **Neutral** if the cross-price elasticity of j with respect to the price of i is zero.

Point vs Arc Elasticities

- We have studied how to compute the point elasticity.
- However, if:
 - We don't have access to the exact demand function OR,
 - We have inaccurate information, i.e., we only have a few points in the schedule OR,
 - Price changes under consideration are large.
- If we have **at least two points on the demand curve**, then we use the **arc-elasticity of demand**.

$$\bar{Q} = \frac{Q_1 + Q_2}{2}$$

$$\bar{P} = \frac{P_1 + P_2}{2}$$

$$\bar{\epsilon}_p^D = \frac{\frac{\Delta Q}{\bar{Q}}}{\frac{\Delta p}{\bar{P}}} = \left(\frac{\Delta Q}{\Delta p} \right) \left(\frac{\bar{P}}{\bar{Q}} \right), = \frac{(Q_2 - Q_1) / \left(\frac{Q_1 + Q_2}{2} \right)}{(P_2 - P_1) / \left(\frac{P_1 + P_2}{2} \right)} = \frac{\frac{Q_2 - Q_1}{Q_1 + Q_2}}{\frac{P_2 - P_1}{P_1 + P_2}}$$

where the bar denotes taking an average.

For this you need two points.
• (Q_1, P_1)
• (Q_2, P_2)

Price Elasticity of supply

Percentage change in quantity supplied of a good resulting from a 1% increase in its **own price**.

$$\epsilon_p^S = \frac{\Delta\%Q}{\Delta\%p} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta p}{p}} = \left(\frac{\Delta Q}{\Delta p} \right) \left(\frac{p}{Q} \right).$$

① ②

For a linear supply curve $Q_s(p) = c + dp$, the above expression simplifies to

$$\epsilon_p^S = d \frac{p}{Q}.$$

① ②

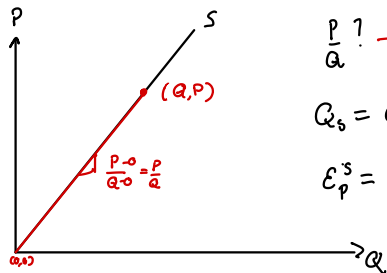
The price elasticity of supply is always positive, since price changes always move in the same direction from changes in quantity supplied (law of supply).

Discussion

- Notice that in the expression for the point-price elasticity of a linear supply curve

$$\epsilon_p^S = d \frac{P}{Q},$$

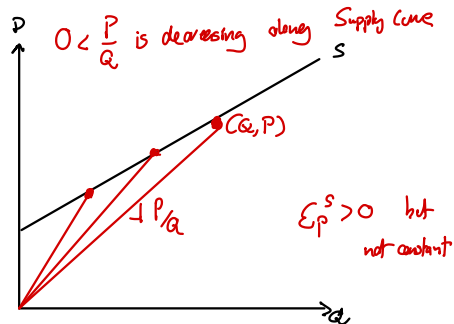
d is constant, but what about the ratio P/Q ?



$\frac{P}{Q} ? \rightarrow$ always
a constant

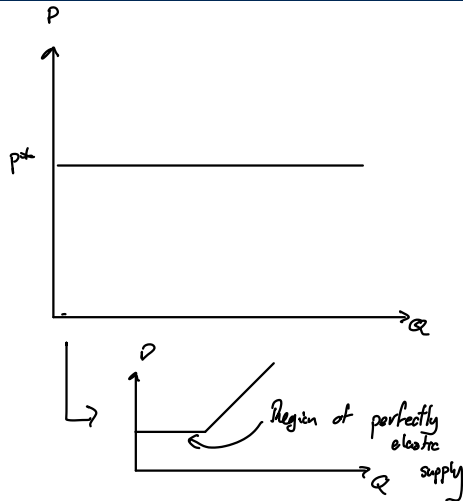
$$Q_S = dP$$

$$\epsilon_p^S = d \cdot \frac{P}{dP} = 1$$



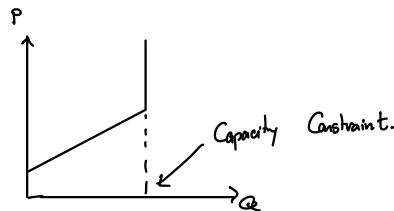
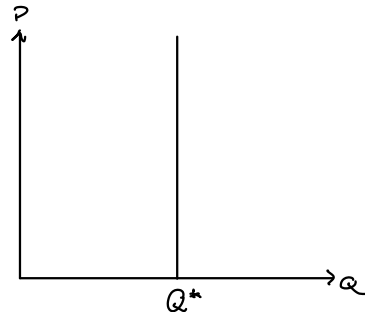
Infinitely Elastic Supply

- For a horizontal supply curve $\Delta Q / \Delta p$ is infinite.
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- Intuition: producers will offer as much of a good as they can at a single price, but for any higher price, quantity supplied increases without limit, while for any lower price the quantity supplied drops to zero.



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Determinants of Supply Elasticity

- Flexibility of inputs.
- Mobility of inputs.
- Ability to produce substitute inputs.
- Time.